Run Payjoin Infrastructure

Deploy lightweight, zero-custody infrastructure that strengthens Bitcoin privacy for everyone

What Is Payjoin Mailroom?

`payjoin-mailroom` is a single, lightweight binary that bundles the two server-side roles required by Async Payjoin (BIP 77):

- **Payjoin Directory**: a store-and-forward mailbox that holds small, ephemeral, end-to-end encrypted payloads so a sender and receiver can complete a payjoin asynchronously (they don't need to be online at the same time).
- **OHTTP Relay**: a simple HTTP proxy that separates client IP addresses from the directory, preventing the directory from correlating users with their network identity. No loopbacks between the relay and directory within the same instance.

Together, these two components are the backbone that makes Async Payjoin work for every wallet.

Why Does This Matter?

Payjoin makes it possible to break Satoshi's original privacy concern: that all inputs to a transaction belong to one person.

With Async Payjoin, receivers no longer need to run a server. But this convenience depends on directory and relay infrastructure existing. Today, only a small handful of operators run these services. A single point of failure is a single point of censorship.

Why Does It Matter Who Runs This?

Bitcoin privacy infrastructure works best when it's operated by entities that can sustain it long-term. We don't need a sprawling network of operators. Fewer directories means a larger anonymity set. The more users share a single directory, the harder traffic analysis becomes. But we do need **a few more resilient operators** for redundancy and resistance to censorship.

Trust Model: YOU SEE NOTHING

All Async Payjoin payloads are end-to-end encrypted with HPKE. The directory cannot read, forge, or correlate transaction contents. The OHTTP relay only sees encrypted blobs and never learns who is paying whom. No loopbacks between relay and directory roles.

The separation also applies to your own wallet. If your organization ships a wallet, do not default it to your own directory when that wallet syncs through your own Electrum, Esplora or mempool backend. The backend already sees the user's IP address. Pointing the same user at your directory hands you their mailbox as well, and the relay in between protects nothing. Default your wallet to another operator's directory and let others default to yours.

You never touch bitcoin. You never see transactions. Zero custody risk.

V2 (BIP 77) payloads are fully end-to-end encrypted. V1 backwards-compatible requests (BIP 78) are plaintext, meaning the directory can see v1 sender IP addresses and transaction contents. V1 support is transitional and will be removed as wallets upgrade to V2. V1 is off unless an operator enables it.

Aggregate Metrics: Four Integers a Week

payjoin-mailroom 0.1.3 reports coarse aggregate metrics to the Payjoin Foundation, and listed operators enable it. Build with the `telemetry` feature and add a `[telemetry]` section to the config. Without both, nothing is exported. Logs are never exported.

What is collected. Four integers a week, each covering the most recently completed Monday-to-Monday UTC week: HTTP requests completed, HTTP requests started, mailbox entries written, and OHTTP bootstrap tunnels refused at the concurrency cap. Nothing else leaves your server.

What is not collected. The data points carry no attributes: no IP address, mailbox ID, hostname, endpoint, method, status code or protocol version. No payload contents or sizes. No logs. Only the settled week is exported, so there is no live value to watch and no overlapping windows to difference for daily traffic. The export is labelled with exactly two attributes: the service name and the domain you choose. Counts are exact, including zero. Rounding would hide at most a few units from a passive viewer, and anyone able to send traffic could add a known amount and subtract it back out.

Why this does not harm privacy. Payjoin privacy rests on nobody being able to tie a user to a mailbox or a transaction. A weekly total across every user of a mailroom, exported after the week has closed, carries no identifier, no timestamp finer than a week and no link to the chain, so it cannot be tied to anyone.

Trust model. Users trust nothing new. Operators keep their per-request logs on their own machine and send the Foundation four integers a week, labelled with their domain. The Foundation receives those series and publishes aggregate dashboards from them. Each rule above is pinned by a unit test in the mailroom source, and the full description is the Telemetry section of the payjoin-mailroom README.

What You Need

REQUIREMENT	DETAILS
Server	A $5/month VPS is sufficient. Minimal CPU and RAM.
Domain + TLS	A domain name with TLS. ACME is built in for automatic certificate provisioning.
Roles	Listed operators run both roles, directory and relay, under one domain.
Storage	Very minimal. Only config details are persisted. Payjoin payloads are ephemeral.
Deployment	Docker image, Nix flake, or build from source. Inline config supported.
Metrics	Build with the `telemetry` feature and fill in the `[telemetry]` section. Four integers a week leave your server. Nothing else does.
Bitcoin Node	**Not required.** Pure relay infrastructure. No chain access needed.

FAQ

Can the operator steal or censor bitcoin?

No. The service never holds keys, constructs transactions, or accesses a bitcoin node. Payloads are encrypted end-to-end. The worst an operator can do is go offline, in which case wallets automatically fall back to the typical (non-payjoin) transaction by using the address included in the Bitcoin URI. No funds are ever at risk.

What is the service comparable to, architecturally?

Think of a CDN or encrypted email relay. You forward opaque, encrypted blobs between parties. You cannot inspect, modify, or correlate the contents. Even a fully compromised operator learns nothing useful.

What about OFAC compliance?

payjoin-mailroom can screen plaintext V1 requests against a blocked-address list, such as the OFAC list, and can filter requests by IP address regardless of protocol version. Async Payjoin sessions are encrypted end to end, so their contents cannot be screened. Operators can enforce their own compliance policies without any changes to the protocol.

What data does the operator see?

Encrypted blobs and their own server logs. The directory sees fixed-size encrypted mailboxes and no IP addresses. The relay sees IP addresses and no content. The operator never sees payload contents, Bitcoin addresses, or sender/receiver identity. The exception is V1 backwards-compatible requests, which are plaintext. V1 is transitional and off unless an operator enables it. What leaves the server is four weekly integers, described under Aggregate Metrics. This is less visibility than even Signal has running their own infrastructure, since Signal registers phone numbers and sees payload sizes. payjoin-mailroom does neither.

Is there a revenue model?

Public-good infrastructure, and it is mutual. Your mailroom serves other wallets' users, and your own wallet's users rely on other operators' mailrooms. Running one is a reputation signal that your organization takes Bitcoin privacy seriously.

Get Started

github.com/payjoin/rust-payjoin/tree/master/payjoin-mailroom

Questions? Reach out at **payjoin.org** or open a Discussion on **GitHub**.
